



COMPUTER
ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	Automating Server Management	SECTION:	CPE31S2
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ACTIVITY NO. ____ Prelim Exam

1. PROCEDURE OUTPUT

Part A

Submit a screenshot of the following:

- Ansible version

```
wdjavillonar@workstation:~$ sudo apt update
[sudo] password for wdjavillonar:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
71 packages can be upgraded. Run 'apt list --upgradable' to see them.
wdjavillonar@workstation:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/wdjavillonar/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/wdjavillonar/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

```
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo apt update
[sudo] password for serv1javillonar:
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
86 packages can be upgraded. Run 'apt list --upgradable' to see them.
serv1javillonar@serv1javillonar-VirtualBox:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/serv1javillonar/.ansible/plugins/module
s', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/serv1javillonar/.ansible/collections:/usr/s
hare/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/py
thon3)
  jinja version = 3.1.2
  libyaml = True
```

```
serv2javillonar@server2:~$ sudo apt update
[sudo] password for serv2javillonar:
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
85 packages can be upgraded. Run 'apt list --upgradable' to see them.
serv2javillonar@server2:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/serv2javillonar/.ansible/plugins/modul
es', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/serv2javillonar/.ansible/collections:/usr/
share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bi
n/python3)
  jinja version = 3.1.2
  libyaml = True
```

- SSH connectivity

```
wdjavillonar@workstation:~$ ssh serv1javillonar@192.168.56.101
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

Expanded Security Maintenance for Applications is not enabled.

92 updates can be applied immediately.

57 of these updates are standard security updates.

To see these additional updates run: `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

Last login: Fri Aug 28 16:14:42 2026 from 192.168.56.103

```
serv1javillonar@serv1javillonar-VirtualBox:~$ exit
```

logout

Connection to 192.168.56.101 closed.

```
wdjavillonar@workstation:~$ ssh serv2javillonar@192.168.56.102
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-30-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

Expanded Security Maintenance for Applications is not enabled.

33 updates can be applied immediately.

To see these additional updates run: `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

Last login: Fri Aug 28 16:15:04 2026 from 192.168.56.103

```
serv2javillonar@server2:~$ exit
```

logout

Connection to 192.168.56.102 closed.

Part B

Demonstrate the following:

- Inventory.yaml

```
wdjavillonar@workstation:~/ansible-lab$ cat inventory.yaml
all:
  children:

    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.101
          ansible_user: serv1javillonar

        ubuntu2:
          ansible_host: 192.168.56.102
          ansible_user: serv2javillonar
```

- Successful inventory verification

```
wdjavillonar@workstation:~/ansible-lab$ ansible all --list-hosts
hosts (3):
  ubuntu1
  ubuntu2
  ubuntu3

wdjavillonar@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --li
st
{
  "_meta": {
    "hostvars": {
      "ubuntu1": {
        "ansible_host": "192.168.56.101"
      },
      "ubuntu2": {
        "ansible_host": "192.168.56.102"
      },
      "ubuntu3": {
        "ansible_host": "192.168.56.103"
      }
    }
  },
  "all": {
    "children": [
      "ungrouped",
      "ubuntu"
    ]
  },
  "ubuntu": {
    "hosts": [
      "ubuntu1",
      "ubuntu2",
      "ubuntu3"
    ]
  }
}
```

- Successful ping results

```
wdjavillonar@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
ubuntu1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

Part C

Demonstrate successful execution of the metagroup.

```
wdjavillonar@workstation:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yaml
ubuntu1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
wdjavillonar@workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.yaml
hosts (2):
  ubuntu1
  ubuntu2
```

Part D

Demonstrate successful use of variables.

```
wdjavillonar@workstation:~/ansible-lab$ cat testvars.yaml
---
- name: Display Variables
  hosts: linuxservers

  tasks:

    - name: Show Operating System
      debug:
        msg: " Host {{ inventory_hostname }} is running {{ os_type }}"

wdjavillonar@workstation:~/ansible-lab$ nano testvars.yaml
wdjavillonar@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] *****

TASK [Gathering Facts] *****
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Show Operating System] *****
ok: [ubuntu1] => {
  "msg": " Host ubuntu1 is running Ubuntu"
}
ok: [ubuntu2] => {
  "msg": " Host ubuntu2 is running Ubuntu"
}

PLAY RECAP *****
ubuntu1          : ok=2    changed=0    unreachable=0    failed=0    skipped=0
  rescued=0      ignored=0
ubuntu2          : ok=2    changed=0    unreachable=0    failed=0    skipped=0
  rescued=0      ignored=0
```

Part E

Demonstrate:

- Successful playbook execution
- Successful task completion on all managed nodes

```
wdjavillonar@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playbook.yaml

PLAY [My first playbook] *****

TASK [Gathering Facts] *****
ok: [ubuntu2]
ok: [ubuntu1]

TASK [Ping all managed nodes] *****
ok: [ubuntu1]
ok: [ubuntu2]

TASK [Display Hello World] *****
ok: [ubuntu1] => {
  "msg": "Hello World from ubuntu1"
}
ok: [ubuntu2] => {
  "msg": "Hello World from ubuntu2"
}

TASK [Display Operating Systems] *****
ok: [ubuntu1] => {
  "msg": "Operating System: Ubuntu"
}
ok: [ubuntu2] => {
  "msg": "Operating System: Ubuntu"
}

PLAY RECAP *****
ubuntu1      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu2      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a new group named **servers** that contains only:

- ubuntu1
- centos1

Create a playbook named **challenge.yaml** that displays the message:

None

Welcome to Ansible Laboratory

Run the playbook and demonstrate that only hosts in the **servers** group receive the message.

```

wdjavillonar@workstation:~/ansible-lab$ nano challenge.yaml
wdjavillonar@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.yaml

PLAY [Supplementary Activity] *****

TASK [Gathering Facts] *****
ok: [ubuntu1]

TASK [Display Welcome Message] *****
ok: [ubuntu1] => {
  "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP *****
ubuntu1                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescue=0
                        : ignored=0

wdjavillonar@workstation:~/ansible-lab$ cat challenge.yaml
---
- name: Supplementary Activity
  hosts: servers

  tasks:

    - name: Display Welcome Message
      debug:
        msg: "Welcome to Ansible Laboratory"

```

Guide Questions

1. What is the role of the Control Node in Ansible?

It is the host machine where Ansible is installed and executed. It reads your playbooks and inventories, connects to the machines using SSH and executes modules to automate tasks.

2. Why is an inventory file required?

It acts as Ansible's address book and it defines what managed nodes Ansible will target, grouping by hosts names or IP addresses so commands and playbooks know where to run.

3. What is the purpose of a metagroup?

It organizes multiple sub-groups into single parent group to allow to apply configuration settings, variables or playbooks to distinct server roles simultaneously without duplicating.

4. How do variables simplify Ansible administration?

It lessens hardcoded values, allowing to reuse same playbooks across different environments by substituting environment parameters like hostnames, passwords, or port numbers.

5. What is the difference between an ad hoc command and a playbook?

The ad hoc is a quick and single line command used for one off tasks like rebooting a server or checking uptime and it is not saved for reuse. While, playbook is a reusable YAML file containing a structured set of instructions, plays, and tasks for complex configuration management.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic



work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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